

Gaurav Waratkar

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I am a Post-Doctoral Fellow at Caltech where I am developing next-generation X-ray detectors for future space missions & building a new infrared telescope ‘Cryoscope’ to be deployed at Antarctica. My work focuses on designing space hardware, building next-generation instruments, and studying fast-transients.

Education

California Institute of Technology

USA

Post-Doctoral Fellow (Space Radiation Lab & Caltech Optical Observatories)

2025 – Present

Host: Fiona Harrison & Mansi Kasliwal

Topics: Developing next-generation X-ray detectors & building infrared telescope ‘Cryoscope’

Indian Institute of Technology Bombay

India

Ph.D. in Astrophysics

2020 – 2025

Advisor: Prof. Varun Bhalerao

Thesis: High Energy Transients: AstroSat-CZTI data analysis & Daksha instrument development

Indian Institute of Technology Bombay

India

B.Tech. in Mechanical Engineering

2015 – 2019

Awards & Fellowships

2026 ASI AstroSat Award; given by Astronomical Society of India (Honorable Mention)

2026 Keck Institute of Space Studies (KISS) Affiliate, nominated by Caltech faculty

2024 LIGO Science Collaboration Fellow; on-site commissioning at LIGO Livingston

Publications & Press Releases

A complete list of all of my publications is — [here on NASA-ADS](#).

You can find a list of published articles on the last page of this document.

Selected Media Coverage.....

2025 IIT Bombay-led Daksha mission, *Times of India*, *WION*, *India Today*

2025 Searching for cosmic bangs and whispers, *Research Matters*

2024 Low-latency data products for gravitational wave mergers, *Phys.org*

2023 India’s Astrosat captures 600th mega explosion, Media coverage across India

2022 India’s AstroSat witnesses black hole birth for the 500th time, Media coverage across India

Selected Talks and Conferences

2025 Low-latency data performance and RAVEN for LVK, ASI 2025, NITR, India

2025 X-ray counterparts of FRBs, Young Astronomers’ Meet, TIFR, India

2024 LVK low-latency data, LIGO Livingston Observatory Seminar, USA

2024 High-energy transients with AstroSat and Daksha, Caltech TDA, USA

2024 Low-latency data products and MMA coincidences, Caltech LIGO, USA

2024 Daksha mission overview, UC Berkeley Space Sciences Laboratory, USA

- 2024 Daksha: High-energy transient alerts, LSU Astrophysics, USA
- 2024 Rapid searches for fast transients using AstroSat-CZTI, Transients 2024, IIT Bombay, India
- 2023 X-ray counterparts to LVK triggers, APRIM 2023, Japan
- 2023 EM counterparts to LVK triggers, Amaldi 15, Online
- 2023 Implications of LIGO-India for EMGW, LIGO-India Science Meeting, IUCAA, India
- 2022 CIFT pipeline for LIGO-Virgo triggers, Young Astronomers' Meet, ARIES Nainital, India
- 2021 Fast transient searches with AstroSat-CZTI, 39th ASI Meeting, Online

Observing Experience

- 2026 80 ks; NuSTAR DDT for GRB 260310A / AT2026fgk; PI
- 2026 40 ks; NuSTAR DDT for GRB 260226A; PI
- 2024 200+ hours with the LIGO-Virgo-KAGRA Rapid Response Team
- 2024 ToO observations of BNS candidate 'S240422ed' with 2.5m PRL Telescope
- 2023 Chandra ToO observations of GRB 230812B; Co-PI
- 2022 AstroSat ToO observations of GRB 220408A; Co-PI
- 2019 Several weeks on operations of GROWTH-India telescope
- 2018 Several days on radio observing campaigns with uGMRT

Mentoring

Mentored 28+ undergraduate and master's students on research projects (at least a semester long) in space instrumentation, transient astrophysics, and data analysis.

- 2025-2026 Abhinav: *Gamma Ray Bursts: Search, Analysis and Localisation*
- 2024-2025 Suryansh: *CIFT2 development for AstroSat-CZTI*
- 2024-2025 Atreyi: *Automated T90 and triggered searches for CIFT2*
- 2022-2025 Meghna: *FRB searches, CIFT1.5 automation*
- 2023-2024 Dhananjay: *Daksha self-localization and IPN with Moonbeam*
- 2023-2024 Puja: *Joint GW-GRB searches with CZTI & RAVEN*
- 2022-2024 Ashwin: *CZTI FAR estimates and TGF searches*
- 2023-2024 Rohan: *Energy-dependent searches in CZTI*
- 2023-2024 Abhishek: *Daksha design branding through IDC*
- 2023-2024 Pal: *Daksha data simulator*
- 2023-2024 Ananya: *Triggered searches in CIFT1.5*
- 2023-2024 Vedagya: *Blind searches for CIFT1.5*
- 2024 Tamojeet: *GIT-EMGW galaxy tiles searches*
- 2022-2023 Arhaan: *Automated Kafka GCN stream DB & searches*
- 2023 Bhuvanambiga: *CIFT1.5 triggered searches*
- 2022-2023 Amrutha: *Daksha thermal analysis*
- 2022-2023 Harsh: *GIT-EMGW galaxy tiled searches*
- 2021-2022 Aswin: *CIFT1.5 blind search automation, injections*
- 2020-2021 Srividya: *Daksha baseplate optimization*

- 2020 Soumyadeep: *ASIMOV: Astronomical Satellites: Is My Object Visible*
- 2020 Aditi: *Daksha structural analysis & vibration tests*
- 2020-2021 Drishika: *CIFT1 GRB searches*

Workshops & Teaching

- 2025 SOC, *Young Astronomer's Meet, TIFR Mumbai, India*
- 2023-2025 Mentor, *Setup of an amateur observatory on the IIT Bombay campus*
- 2024 Mentor, *Offline Study Hub (30 students), LVK Gravitational Wave Open Data Workshop*
- 2024 Section Lead, *Hands-on tutorial on Daksha mock data, ASI Workshop*
- 2023 Mentor, *Offline Study Hub (30 students), LVK Gravitational Wave Open Data Workshop*
- 2020 Session Lead, *EMGW Workshop (30 students), Techfest IIT Bombay*
- 2018 Tutor, *GROWTH Winter School*
- 2022-2024 Teaching Assistant, *PH 119 – Physics Laboratory (5 semesters)*
- 2020-2021 Teaching Assistant, *PH 556 – Astrophysics (2 semesters)*

Outreach & Volunteering

- 2026 Volunteer, *Night of Ideas, Wende Museum / French Embassy, Caltech Astro Outreach*
- 2025 Volunteer, *California Dark Sky Festival, Caltech Astro Outreach*
- 2025 Lead, *Sitaron Mein Safar: Interactions with space industry leaders in India*
- 2025 Speaker, *Engineering in Astrophysics, Physics X Mechanical Journal Club, IIT Bombay*
- 2025 Panelist, *Careers in Astronomy, Krittika - IIT Bombay*
- 2019-2025 Mentor, *Krittika - The Astronomy Club of IIT Bombay*
- 2022–2025 Social Media Lead, *Daksha Mission Outreach (Instagram, X, Facebook)*
- 2022–2024 Booth Coordinator, *Daksha – TechConnect, TechFest IIT Bombay*
- 2017–2024 Outreach, *LIGO-India, TechFest IIT Bombay*
- 2018 LOC, *Transients 2024 Conference, IIT Bombay*
- 2023–2024 Booth Coordinator, *LIGO-India Outreach, Technovanza, VJTI Mumbai*
- 2024 Panelist, *Careers in Astronomy, Krittika - IIT Bombay*
- 2024 Invited Talk, *Astrophysics Adventures: AstroSat, LIGO-India, ZetaGravity X-Space*
- 2023 Invited Talk, *Cosmic Exotic Marvels, Pawar International School, Mumbai*
- 2022–2023 Convener, *Astro-Cosmo-Gravity Journal Club, IIT Bombay*
- 2021–2022 Institute Coordinator, *Codetantra (B2B EdTech SaaS), IIT Bombay*
- 2022 LOC, *Astronomers for Planet Earth International Symposium*
- 2021 Climate Reality Leader, *Climate Reality Project*
- 2020 Speaker, *Basics of Astronomical Telescopes, Krittika Lecture Series*
- 2018 Speaker, *Pulsars and Radio Detection Techniques, Krittika Lecture Series*
- 2018 LOC, *GROWTH Annual Conference & Winter School*
- 2016–2018 Manager and Convener, *Krittika – The Astronomy Club of IIT Bombay*
- 2016 Runner-up, *Hostel 7 - IIT Bombay Volleyball GC*

Refereed Papers

The following is the list of the papers that I have co-authored (including the ones in-review / pre-print).

1. The LIGO Scientific Collaboration, the Virgo Collaboration, the KAGRA Collaboration, et al. (2026), arXiv e-prints, *Narrowband searches for continuous gravitational waves from known pulsars in the first two parts of the fourth LIGO–Virgo–KAGRA observing run*, arXiv:2603.25938.
2. The LIGO Scientific Collaboration, the Virgo Collaboration, the KAGRA Collaboration, et al. (2026), arXiv e-prints, *Searches for Continuous Gravitational Waves from Supernova Remnants in the first part of the LIGO–Virgo–KAGRA Fourth Observing run*, arXiv:2603.25808.
3. The LIGO Scientific Collaboration, the Virgo Collaboration, the KAGRA Collaboration, et al. (2026), arXiv e-prints, *GWTC-4.0: Tests of General Relativity. III. Tests of the Remnants*, arXiv:2603.19021.
4. The LIGO Scientific Collaboration, the Virgo Collaboration, the KAGRA Collaboration, et al. (2026), arXiv e-prints, *GWTC-4.0: Tests of General Relativity. II. Parameterized Tests*, arXiv:2603.19020.
5. The LIGO Scientific Collaboration, the Virgo Collaboration, the KAGRA Collaboration, et al. (2026), arXiv e-prints, *GWTC-4.0: Tests of General Relativity. I. Overview and General Tests*, arXiv:2603.19019.
6. The LIGO Scientific Collaboration, the Virgo Collaboration, the KAGRA Collaboration, et al. (2026), arXiv e-prints, *All-sky Searches for Continuous Gravitational Waves from Isolated Neutron Stars in the Data from the First Part of the Fourth LIGO–Virgo–KAGRA Observing Run*, arXiv:2603.14168.
7. Swain, V., Bhalerao, V., Kumar, H., et al. (2026), The Astrophysical Journal, *GRB 230204B: GIT Discovery of a Fast Fading Afterglow Associated with an Energetic Gamma-Ray Burst from a Massive Star Progenitor*, 1000, 58.
8. The IceCube Collaboration, Abbasi, R., Ackermann, M., et al. (2026), arXiv e-prints, *Deep Search for Joint Sources of Gravitational Waves and High-Energy Neutrinos with IceCube During the Third Observing Run of LIGO and Virgo*, arXiv:2601.07595.
9. Abac, A. G., Abouelfettouh, I., Acernese, F., et al. (2026), Physical Review Letters, *Black Hole Spectroscopy and Tests of General Relativity with GW250114*, 136, 041403.
10. Eappachen, D., Balasubramanian, A., Swain, V., et al. (2026), Monthly Notices of the Royal Astronomical Society, *Characterizing EP241107a: multiwavelength observations of an Einstein Probe-detected fast X-ray transient*, 545, staf2062.
11. The LIGO Scientific Collaboration, the Virgo Collaboration, the KAGRA Collaboration, et al. (2025), arXiv e-prints, *Constraints on gravitational waves from the 2024 Vela pulsar glitch*, arXiv:2512.17990.
12. The LIGO Scientific Collaboration, the Virgo Collaboration, the KAGRA Collaboration, et al. (2025), arXiv e-prints, *GWTC-4.0: Searches for Gravitational-Wave Lensing Signatures*, arXiv:2512.16347.
13. Abac, A. G., Abouelfettouh, I., Acernese, F., et al. (2025), The Astrophysical Journal Letters, *GWTC-4.0: An Introduction to Version 4.0 of the Gravitational-Wave Transient Catalog*, 995, L18.
14. The LIGO Scientific Collaboration, the Virgo Collaboration, the KAGRA Collaboration, et al. (2025), arXiv e-prints, *Search for planetary-mass ultra-compact binaries using data from the first part of the LIGO–Virgo–KAGRA fourth observing run*, arXiv:2511.19911.
15. The LIGO Scientific Collaboration, the Virgo Collaboration, the KAGRA Collaboration, et al. (2025), arXiv e-prints, *All-sky search for continuous gravitational-wave signals from unknown neutron stars in binary systems in the first part of the fourth LIGO–Virgo–KAGRA observing run*, arXiv:2511.16863.
16. Abac, A. G., Abouelfettouh, I., Acernese, F., et al. (2025), Physical Review D, *All-sky search for short gravitational-wave bursts in the first part of the fourth LIGO–Virgo–KAGRA observing run*, 112, 102005.

17. Arya, A., Bilkhu, H. S., Vishwakarma, S., et al. (2025), *Journal of Astrophysics and Astronomy, Development of a Compton imager setup*, 46, 84.
18. Abac, A. G., Abouelfettouh, I., Acernese, F., et al. (2025), *The Astrophysical Journal Letters, GW231123: A Binary Black Hole Merger with Total Mass $190\pm 65 M_{\odot}$* , 993, L25.
19. Abac, A. G., Abouelfettouh, I., Acernese, F., et al. (2025), *The Astrophysical Journal Letters, GW241011 and GW241110: Exploring Binary Formation and Fundamental Physics with Asymmetric, High-spin Black Hole Coalescences*, 993, L21.
20. The LIGO Scientific Collaboration, the Virgo Collaboration, the KAGRA Collaboration, et al. (2025), arXiv e-prints, *Direct multi-model dark-matter search with gravitational-wave interferometers using data from the first part of the fourth LIGO-Virgo-KAGRA observing run*, arXiv:2510.27022.
21. The LIGO Scientific Collaboration, the Virgo Collaboration, the KAGRA Collaboration, et al. (2025), arXiv e-prints, *Cosmological and High Energy Physics implications from gravitational-wave background searches in LIGO-Virgo-KAGRA's O1-O4a runs*, arXiv:2510.26848.
22. The LIGO Scientific Collaboration, the Virgo Collaboration, the KAGRA Collaboration, et al. (2025), arXiv e-prints, *Directional Search for Persistent Gravitational Waves: Results from the First Part of LIGO-Virgo-KAGRA's Fourth Observing Run*, arXiv:2510.17487.
23. The LIGO Scientific Collaboration, the Virgo Collaboration, the KAGRA Collaboration, et al. (2025), arXiv e-prints, *Directed searches for gravitational waves from ultralight vector boson clouds around merger remnant and galactic black holes during the first part of the fourth LIGO-Virgo-KAGRA observing run*, arXiv:2509.07352.
24. The LIGO Scientific Collaboration, the Virgo Collaboration, the KAGRA Collaboration, et al. (2025), arXiv e-prints, *GWTC-4.0: Constraints on the Cosmic Expansion Rate and Modified Gravitational-wave Propagation*, arXiv:2509.04348.
25. Abac, A. G., Abouelfettouh, I., Acernese, F., et al. (2025), *Physical Review Letters, GW250114: Testing Hawking's Area Law and the Kerr Nature of Black Holes*, 135, 111403.
26. Roychowdhury, T., Choudhary, H., Bhalariao, V., et al. (2025), *Journal of Astrophysics and Astronomy, Efficacy of galaxy catalogues for following up gravitational wave events*, 46, 68.
27. The LIGO Scientific Collaboration, the Virgo Collaboration, the KAGRA Collaboration, et al. (2025), arXiv e-prints, *Upper Limits on the Isotropic Gravitational-Wave Background from the first part of LIGO, Virgo, and KAGRA's fourth Observing Run*, arXiv:2508.20721.
28. The LIGO Scientific Collaboration, the Virgo Collaboration, the KAGRA Collaboration, et al. (2025), arXiv e-prints, *GWTC-4.0: Population Properties of Merging Compact Binaries*, arXiv:2508.18083.
29. The LIGO Scientific Collaboration, the Virgo Collaboration, the KAGRA Collaboration, et al. (2025), arXiv e-prints, *GWTC-4.0: Updating the Gravitational-Wave Transient Catalog with Observations from the First Part of the Fourth LIGO-Virgo-KAGRA Observing Run*, arXiv:2508.18082.
30. The LIGO Scientific Collaboration, the Virgo Collaboration, the KAGRA Collaboration, et al. (2025), arXiv e-prints, *GWTC-4.0: Methods for Identifying and Characterizing Gravitational-wave Transients*, arXiv:2508.18081.
31. The LIGO Scientific Collaboration, the Virgo Collaboration, the KAGRA Collaboration, et al. (2025), arXiv e-prints, *Open Data from LIGO, Virgo, and KAGRA through the First Part of the Fourth Observing Run*, arXiv:2508.18079.
32. Waratkar, G., Dixit, M., Tendulkar, S. P., et al. (2025), *Journal of Astrophysics and Astronomy, AstroSat-CZTI searches for hard X-ray prompt emission from fast radio bursts*, 46, 59.
33. The LIGO Scientific Collaboration, the Virgo Collaboration, the KAGRA Collaboration, et al. (2025), arXiv e-prints, *All-sky search for long-duration gravitational-wave transients in the first part of the fourth LIGO-Virgo-KAGRA Observing run*, arXiv:2507.12282.

34. Saraogi, D., Bala, S., Joshi, J., et al. (2025), Journal of Astrophysics and Astronomy, *Investigating polarization characteristics of GRB 200503A and GRB 201009A*, 46, 38.
35. Abac, A. G., Abbott, R., Abouelfettouh, I., et al. (2025), The Astrophysical Journal, *Search for Gravitational Waves Emitted from SN 2023ixf*, 985, 183.
36. Afroz, S., Agarwalla, S. K., Bhattacharya, D., et al. (2025), arXiv e-prints, *A White Paper on The Multi-Messenger Science Landscape in India*, arXiv:2505.24408.
37. Abac, A. G., Abbott, R., Abouelfettouh, I., et al. (2025), The Astrophysical Journal, *Search for Continuous Gravitational Waves from Known Pulsars in the First Part of the Fourth LIGO-Virgo-KAGRA Observing Run*, 983, 99.
38. Srinivasaragavan, G. P., Perley, D. A., Ho, A. Y. Q., et al. (2025), Monthly Notices of the Royal Astronomical Society, *Multiwavelength analysis of AT 2023sva: a luminous orphan afterglow with evidence for a structured jet*, 538, 351.
39. Perley, D. A., Ho, A. Y. Q., Fausnaugh, M., et al. (2025), Monthly Notices of the Royal Astronomical Society, *The luminous, slow-rising orphan afterglow AT2019pim as a candidate moderately relativistic outflow*, 537, 2362.
40. Burns, E., Fryer, C. L., Agullo, I., et al. (2025), arXiv e-prints, *Multidisciplinary Science in the Multimessenger Era*, arXiv:2502.03577.
41. Raman, G., Ronchini, S., Delaunay, J., et al. (2025), The Astrophysical Journal, *Swift-BAT GUANO Follow-up of Gravitational-wave Triggers in the Third LIGO-Virgo-KAGRA Observing Run*, 980, 207.
42. Abac, A. G., Abbott, R., Abouelfettouh, I., et al. (2024), The Astrophysical Journal, *A Search Using GEO600 for Gravitational Waves Coincident with Fast Radio Bursts from SGR 1935+2154*, 977, 255.
43. Ahumada, T., Anand, S., Coughlin, M. W., et al. (2024), Publications of the Astronomical Society of the Pacific, *Searching for Gravitational Wave Optical Counterparts with the Zwicky Transient Facility: Summary of O4a*, 136, 114201.
44. Waratkar, G., Bhalerao, V., & Bhattacharya, D. (2024), The Astrophysical Journal, *Bright in the Black: Searching for Electromagnetic Counterparts to Gravitational-wave Candidates in LIGO-Virgo-KAGRA Observation Runs with AstroSat-CZTI*, 976, 123.
45. Abac, A. G., Abbott, R., Abe, H., et al. (2024), Physical Review D, *Ultralight vector dark matter search using data from the KAGRA O3GK run*, 110, 042001.
46. Abac, A. G., Abbott, R., Abouelfettouh, I., et al. (2024), The Astrophysical Journal Letters, *Observation of Gravitational Waves from the Coalescence of a $2.54.5 M_{\text{CHANDRA}}$ Compact Object and a Neutron Star*, 970, L34.
47. Bhalerao, V., Vadawale, S., Tendulkar, S., et al. (2024), Experimental Astronomy, *Daksha: on alert for high energy transients*, 57, 24.
48. Bhalerao, V., Sawant, D., Pai, A., et al. (2024), Experimental Astronomy, *Science with the Daksha high energy transients mission*, 57, 23.
49. Saraogi, D., Aditya, J. V., Bhalerao, V., et al. (2024), Monthly Notices of the Royal Astronomical Society, *Localization of gamma-ray bursts using AstroSat Mass Model*, 530, 1386.
50. Chaudhary, S. S., Toivonen, A., Waratkar, G., et al. (2024), Proceedings of the National Academy of Science, *Low-latency gravitational wave alert products and their performance at the time of the fourth LIGO-Virgo-KAGRA observing run*, 121, e2316474121.
51. Bhattacharjee, S., Banerjee, S., Bhalerao, V., et al. (2024), Monthly Notices of the Royal Astronomical Society, *Joint gravitational wave-short GRB detection of binary neutron star mergers with existing and future facilities*, 528, 4255.

52. Ewing, B., Huxford, R., Singh, D., et al. (2024), Physical Review D, *Performance of the low-latency GstLAL inspiral search towards LIGO, Virgo, and KAGRA's fourth observing run*, 109, 042008.
53. Bala, S., Mate, S., Mehla, A., et al. (2023), Journal of Astronomical Telescopes, Instruments, and Systems, *Prospects of measuring gamma-ray burst polarization with the Daksha mission*, 9, 048002.
54. Wagner, S. M., Mingo, B., Majidi, F. Z., et al. (2023), Nature Astronomy, *A more sustainable future for astronomy*, 7, 244.
55. Singhal, A., Palit, S., Bala, S., et al. (2022), Journal of Astrophysics and Astronomy, *Gravitational waves and electromagnetic transients*, 43, 53.
56. Kumar, H., Bhalerao, V., Anupama, G. C., et al. (2022), Monthly Notices of the Royal Astronomical Society, *GROWTH on S190426c II: GROWTH-India Telescope search for an optical counterpart with a custom image reduction and candidate vetting pipeline*, 516, 4517.
57. Kumar, H., Bhalerao, V., Anupama, G. C., et al. (2022), The Astronomical Journal, *India's First Robotic Eye for Time-domain Astrophysics: The GROWTH-India Telescope*, 164, 90.
58. Sharma, Y., Marathe, A., Bhalerao, V., et al. (2021), Journal of Astrophysics and Astronomy, *The search for fast transients with CZTI*, 42, 73.
59. Kasliwal, M. M., Anand, S., Ahumada, T., et al. (2020), The Astrophysical Journal, *Kilonova Luminosity Function Constraints Based on Zwicky Transient Facility Searches for 13 Neutron Star Merger Triggers during O3*, 905, 145.
60. Coughlin, M. W., Ahumada, T., Anand, S., et al. (2019), The Astrophysical Journal Letters, *GROWTH on S190425z: Searching Thousands of Square Degrees to Identify an Optical or Infrared Counterpart to a Binary Neutron Star Merger with the Zwicky Transient Facility and Palomar Gattini-IR*, 885, L19.